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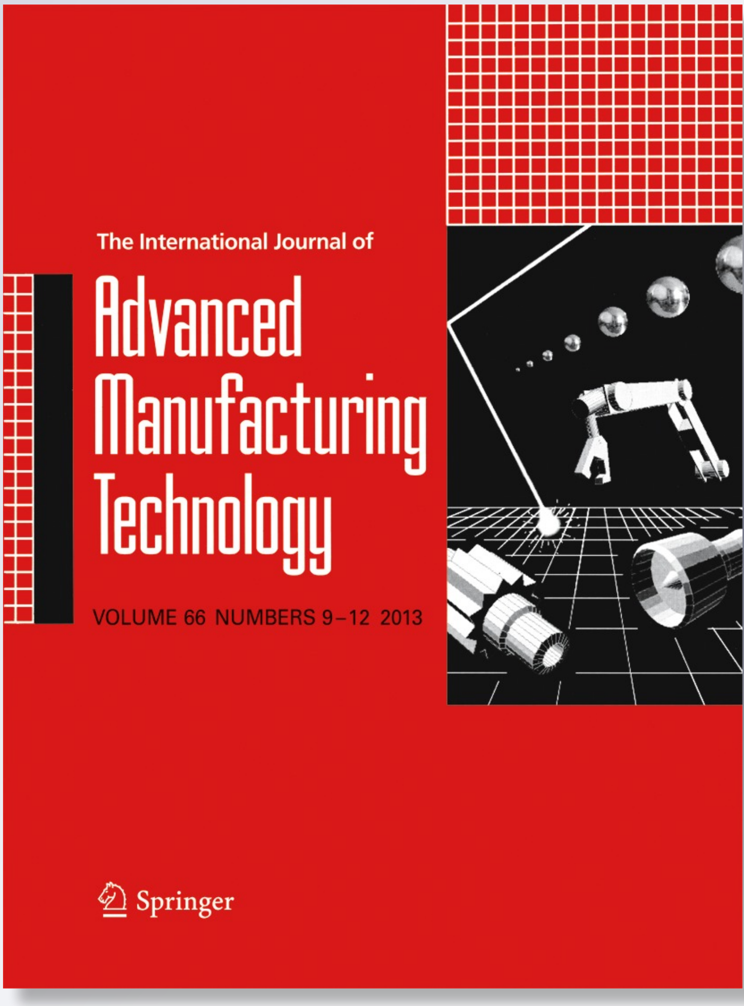

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ORIGINAL ARTICLE

A postprocessor for table-tilting type five-axis machine tool based on generalized kinematics with variable feedrate implementation

Yaman Boz · Ismail Lazoglu

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Abstract Improvements in the machine tool and the machining process technologies increased the need for generic postprocessors in order to exploit the capabilities of the machine tools. Contrary to conventional machining approach, next-generation machining technologies such as force-based feedrate scheduling and toolpath optimization requires the implementation of the variable feedrate during toolpath which constitutes the aim of this article. Therefore, this paper introduces a postprocessor for table-tilting type five-axis machine tool based on generalized kinematics with variable feedrate implementation. Furthermore, a practical yet effective method for avoiding kinematic singularities by spherical interpolation and NC data correction is presented as well. Proposed approach is validated for various five-axis machine tools with different kinematic configurations via virtual machine simulation module. Results of the verification tests show that presented postprocessing approach can accurately convert the cutter location information into NC codes and it is demonstrated that integrated virtual simulation module can simulate toolpaths with large number of blocks.

Keywords Postprocessor · Inverse kinematics · Five-axis machining · NC programming · CNC · CAM

1 Introduction

Five-axis machining is used in aerospace, die-mold, tool making and biomedical industries very extensively. As five-axis machining gained much attention in industry, meanwhile kinematics of five-axis machine tools increased its importance. Five-axis machine tools vary in their kinematic configurations from three-axis machine tools due to their rotary axes; consequently the development of five-axis postprocessors is a challenging task.

Kinematics of five-axis machine tools for different types of configurations is investigated by many authors. One of the first studies on five-axis postprocessing was done by Takeuchi and Watanabe [1]. In this study, a five-axis toolpath generation strategy was presented and obtained CL data was converted into NC code using developed postprocessor. Sakamoto and Inasaki [2] analyzed the kinematics of five-axis machining centers for possible configurations and a form-shaping function was introduced in order to define the kinematic behavior of the machine tool. In the postprocessor development of five-axis machine tools, noteworthy research was conducted by She and Lee [3, 4] where postprocessors for table-tilting, spindle tilting, and table/spindle tilting machine tools were developed. Makhanov [5] also developed kinematic models for different types of machine tools having different kinematical structure. More practically, Bohez [6] investigated important geometric and kinematic properties of five-axis machine tools such as workspace utilization factor, machine tool space efficiency by modeling the machine tool kinematics. Following these studies, She and Huang [7] developed postprocessor for five-axis machine tool with nutating head and table configuration. Similarly, Sørby [8] examined inverse kinematics of a five-axis machine tool with nutating table configuration and proposed an approach for avoiding kinematic singularities. Most recently, Lee and Lin [9] proposed a universal environment for simulating five-axis virtual machine tool and a

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postprocessing method based on the Denavit–Hartenberg notation.

In practice, validity of a postprocessor can be proved by several trial runs on the machine tool. However, using a multi-axis machine for proving out the validity of an NC data is a time-consuming and a risky operation. Therefore, the demand for virtual machine simulation is increasing gradually. On the other hand, commercial virtual machining simulation software may not always be affordable. Furthermore, an integrated virtual machine simulation for postprocessor development may also be timesaving and easy to eliminate errors.

In this article, a generic postprocessor for table-tilting type five-axis machine tool with variable feedrate implementation and a practical method for avoiding kinematic singularities are presented. A virtual machine simulation is developed for the verification of the NC code.

2 Kinematic modeling of five-axis machine tool

Solid model of a BC type table-tilting five-axis milling machine is shown in Fig. 1. The machine has three orthogonal translational axes and two orthogonal rotary axes. The rotary axes B and C are on the table of the machine (Fig. 1.).

A five-axis machine tool can be considered as a kinematic chain with serially linked rigid bodies comprising of three translational and two revolute joints. Therefore, the forward and inverse kinematics of the machine tool can be obtained using homogenous transformation matrices by defining necessary coordinate frames. General forms of transformation matrices for translational and rotational joints are given in matrices (1) and (2) respectively.

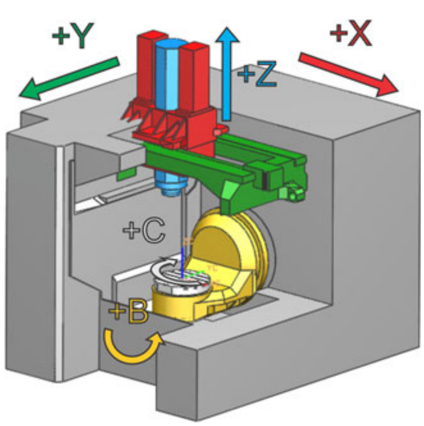


Fig. 1 Solid model of five-axis milling machine

$$T = \begin{bmatrix} 1 & 0 & 0 & x \\ 0 & 1 & 0 & y \\ 0 & 0 & 1 & z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

$$R(V,\theta) = \begin{bmatrix} V_x^2(1-c\theta)+c\theta & V_xV_y(1-c\theta)-V_z\sin\theta & V_xV_z(1-c\theta)+V_ys\theta & 0 \\ V_xV_y(1-c\theta)+V_zs\theta & V_y^2(1-c\theta)+\cos\theta & V_yV_z(1-c\theta)-V_xs\theta & 0 \\ V_xV_z(1-c\theta)-V_ys\theta & V_yV_z(1-c\theta)+V_xs\theta & V_z^2(1-c\theta)+c\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

where $c\theta{=}\cos\theta$ and $s\theta{=}\sin\theta$

Translation matrix T defines a motion along an arbitrary direction, where in vector notation is equal to $T = xi + yj + zk$. The matrix given in Eq. (2) describes a rotation around an arbitrary vector $V = V_xi + V_yj + V_zk$ by an angle of θ. Considering positive direction of rotation according to right hand rule rotation vectors for B and C axes are given as:

$$V_B = [0, -1, 0] \quad (3)$$

$$V_C = [0, 0, -1] \quad (4)$$

For a general five-axis motion, the path between two cutter location (CL) points can be defined as a 3D spline, where cutting tool moves relative to the workpiece. In order to define five-axis motion three coordinate frames will be introduced as:

X_T–Y_T–Z_T: The fixed coordinate frame attached to the tooltip of the cutter.

X_R–Y_R–Z_R: The center of rotation of two rotary axes intersects at a point where machine rotary zero and the rotary zero coordinate frame are located.

X_W–Y_W–Z_W: Workpiece coordinate frame attached to the blank workpiece mounted on the machine. In practical use, it is also called work offset or fixture offset.

Detailed illustration of the coordinate frames is shown in Fig. 2. The vector $\boldsymbol{W}$ in Fig. 2 represents the offset distance from the workpiece coordinate frame to the rotary zero coordinate frames and denoted as $W = W_xi + W_yj + W_zk$.

As it is stated before, the position and the orientation of the cutting tool relative to workpiece coordinate frame can be obtained using homogenous transformations sequentially. First, workpiece coordinate frame is translated by the offset vector $\boldsymbol{W}$. Then, two consecutive rotations for C and B axis of the machine around Z_R and Y_R coordinate frame axes are applied respectively. Finally, the translation by a vector $\boldsymbol{D}$ is applied to construct the kinematic relationship. Overall transformation which

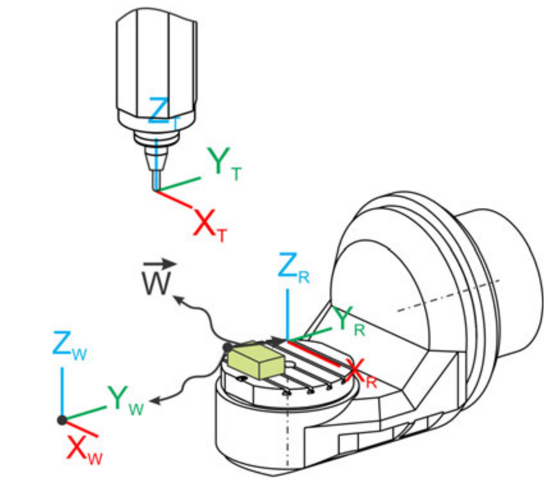


Fig. 2 Definition of coordinate frames

constitutes the forward kinematics of the machine can be given as:

$$\begin{bmatrix} P & T \\ 1 & 0 \end{bmatrix} = T(W_x, W_y, W_z) R(Z_R, -\theta_z) R(Y_R, -\theta_y) T(D_x, D_y, D_z) \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \tag{5}$$

In Eq. (5), θ_z represents the rotation angle of C axis and θ_y represents the rotation angle of B axis of the machine. These two rotations formulated to be in the negative direction because positive angles for machine rotary axes are in the opposite direction.

Inverse kinematics equations can be obtained by solving Eq. (5). In a cutter location source file or simply CL file, the position of the tooltip and the orientation of the tool are given as:

$$P = P_x i + P_y j + P_z k \tag{6}$$

$$T = T_x i + T_y j + T_z k \tag{7}$$

In matrix form, it can be expressed as:

$$\begin{bmatrix} P & T \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} P_x & T_x \\ P_y & T_y \\ P_z & T_z \\ 1 & 0 \end{bmatrix} \tag{8}$$

Since Eqs. (5) and (8) are equal by solving these equations simultaneously, unknown parameters θ_y, θ_z, and $\boldsymbol{D}$ can be found.

$$\begin{bmatrix} T_x \\ T_y \\ T_z \\ 0 \end{bmatrix} = \begin{bmatrix} -c\theta_z s\theta_y \\ s\theta_z s\theta_y \\ c\theta_y \\ 0 \end{bmatrix} \tag{9}$$

$$\begin{bmatrix} P_x \\ P_y \\ P_z \\ 1 \end{bmatrix} = \begin{bmatrix} D_x\{c\theta_z c\theta_y\} + D_y\{s\theta_z\} + D_z\{-c\theta_z s\theta_y\} + W_x \\ D_x\{-s\theta_z c\theta_z\} + D_y\{c\theta_z\} + D_z\{s\theta_z s\theta_y\} + W_y \\ D_x\{s\theta_y\} + D_z\{c\theta_y\} + W_z \\ 1 \end{bmatrix} \tag{10}$$

The rotation angles θ_y and θ_z have to be determined first from Eq. (9). However, rotation angle θ_y is determined and then θ_z is solved accordingly because the solution for θ_z depends on the angle θ_y.

There are two possible solutions for θ_y which can be given as:

$$B = \theta_y = \cos^{-1}(T_z)\, for\, 0 \leq \theta_y \leq \pi \tag{11}$$

$$B = \theta_y = -\cos^{-1}(T_z) \text{ for } -\pi \leq \theta_y \leq 0 \tag{12}$$

Considering the working range of the machine, appropriate solution for B axis is selected and validity of the solution is checked at each CL point. This selection procedure is explained in Section 3 in detail. Rotation angle for C axis is found as:

$$C = \theta_z = atan2(T_y, T_x) \tag{13}$$

where *atan2* is a four quadrant arctangent function.

Once rotation angles of the rotary angles are determined, position of the tooltip denoted as *X, Y, Z* is calculated by solving Eq. (10) simultaneously as:

$$\begin{aligned} X &= D_x + W_x = [P_x - W_x]\{c\theta_z c\theta_y\} + [P_y - W_y]\{-s\theta_z c\theta_y\} \\ &\quad + [P_z - W_z]\{-s\theta_y\} + W_x \\ Y &= D_y + W_y = [P_x - W_x]\{s\theta_z\} + [P_y - W_y]\{c\theta_z\} + W_y \\ Z &= D_z + W_z = [P_x - W_x]\{c\theta_z s\theta_y\} + [P_y - W_y]\{-s\theta_z s\theta_y\} \\ &\quad + [P_z - W_z]\{c\theta_y\} + W_z \end{aligned} \tag{14}$$

3 Linearization and NC data correction

3.1 Linearization

In CAM packages, CL data is generated directly from the toolpath data considering that actual five-axis motion follows a linear trajectory. On the other hand, due to simultaneous movement of linear and rotary axes of the machine, actual toolpath and programmed toolpath between two consecutive CL points deviate. This phenomenon is shown in Fig. 3.

In Fig. 3, points P_i and P_{i+1} are two consecutive CL points from CL data, point P_{int} is the interpolated CL point as a result of linearization. Since, presented post-processor will be used for feedrate scheduling or optimization purposes in five-axis machining; in this study linearization of the generated CL data is utilized because CL data and the NC data have to match each other exactly. Any difference in the CL data and the corresponding NC data will directly affect engagement boundaries of the tool with the workpiece and the resulting cutting force.

Most of the existing interpolation methods [5, 6] are linear and interpolation of tool orientation vectors is not accurate in these approaches. Here a spherical interpolation scheme for more accurate interpolation of the tool axis vectors is introduced.

In five-axis machining, the tool can rotate as well as translate as shown in Fig. 4. Therefore, during translation tool axis rotates from $\overrightarrow{T_i}$ to $\overrightarrow{T_{i+1}}$ around an arbitrary axis $\overrightarrow{k_i}$ an amount of $\Delta\theta$ where rotation axis is both orthogonal to $\overrightarrow{T_i}$ and $\overrightarrow{T_{i+1}}$. Rotation axis $\overrightarrow{k_i}$ and rotation angle $\Delta\theta$ are calculated as,

$$\overrightarrow{k_i} = \begin{Bmatrix} k_x \\ k_y \\ k_z \end{Bmatrix} = \frac{\overrightarrow{T_i} \times \overrightarrow{T}_{i+1}}{\left\| \overrightarrow{T}_i \times \overrightarrow{T}_{i+1} \right\|}$$
$$\Delta\theta = atan2\left(\frac{\left\| \overrightarrow{T}_i \times \overrightarrow{T}_{i+1} \right\|}{\overrightarrow{T_i} \cdot \overrightarrow{T}_{i+1}} \right) \tag{15}$$

Considering the translation and rotation in the tool axis, cutter location points are interpolated linearly and the orientation vectors are interpolated spherically where m changes from 0 to 1. Dividing m to several intervals provides the interpolated CL data.

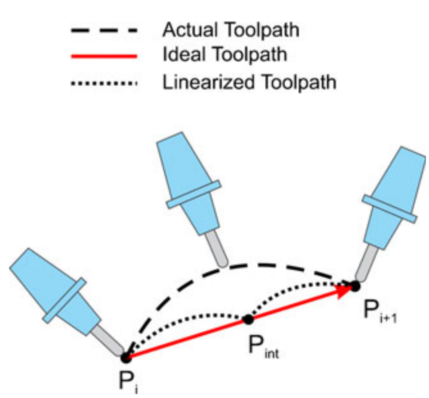


Fig. 3 Linearization in five-axis machining

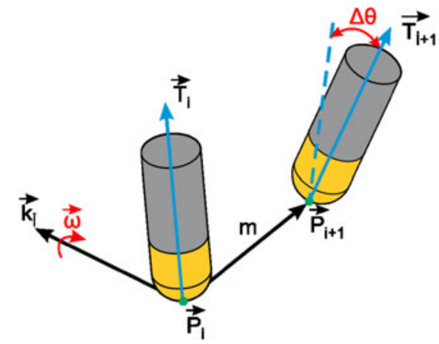


Fig. 4 Tool movement and vectors

$$\begin{aligned} \overrightarrow{P} &= \left(\overrightarrow{P}_{i+1} - \overrightarrow{P}_i \right).m + \overrightarrow{P}_i \\ \overrightarrow{T} &= [R]\overrightarrow{T}_i \end{aligned} \tag{16}$$

The spherical interpolation matrix is given in Eq. 17.

$$[R] = \begin{bmatrix} k_x^2[1-h]+h & k_xk_y[1-h]-k_zg & k_xk_z[1-h]+k_yg \\ k_xk_y[1-h]+k_zg & k_y^2[1-h]+h & k_yk_z[1-h]-k_zg \\ k_xk_z[1-h]-k_yg & k_yk_z[1-h]+k_zg & k_z^2[1-h]+h \end{bmatrix}$$
$$g = \sin(\Delta\theta m) \quad h = \cos(\Delta\theta m) \tag{17}$$

Spherical interpolation is implemented using an example CL data. Original and the interpolated cutter location points and the tool axis vectors are shown in Fig. 5.

3.2 NC data correction

In Section 2, kinematic modeling of the machine tool is introduced and it is shown that there are multiple possible solutions for B and C axis angles. Therefore, appropriate solution among these candidates has to be chosen. For determining the B axis angle a simple yet effective method is used. For the first CL point in the CL data, lead angle of the tool with respect to workpiece is found, and then the solution having the same sign with the lead angle is chosen.

In Fig. 6, the relationship between lead angle and the B axis angle is shown. Lead angle is defined as the rotation angle about Y_w which is Y axis of workpiece coordinate frame. Lead angle is calculated as:

$$\text{lead} = atan2\left(T_x, \sqrt{T_y^2 + T_z^2}\right) \tag{18}$$

For first CL point modifying the solution for B yields:

$$B = \text{sgn}(lead)\cos^{-1}(T_z) \tag{19}$$

Following angles for B axis are determined according to previous B axis angle selecting the solution which causes the shortest rotational motion.

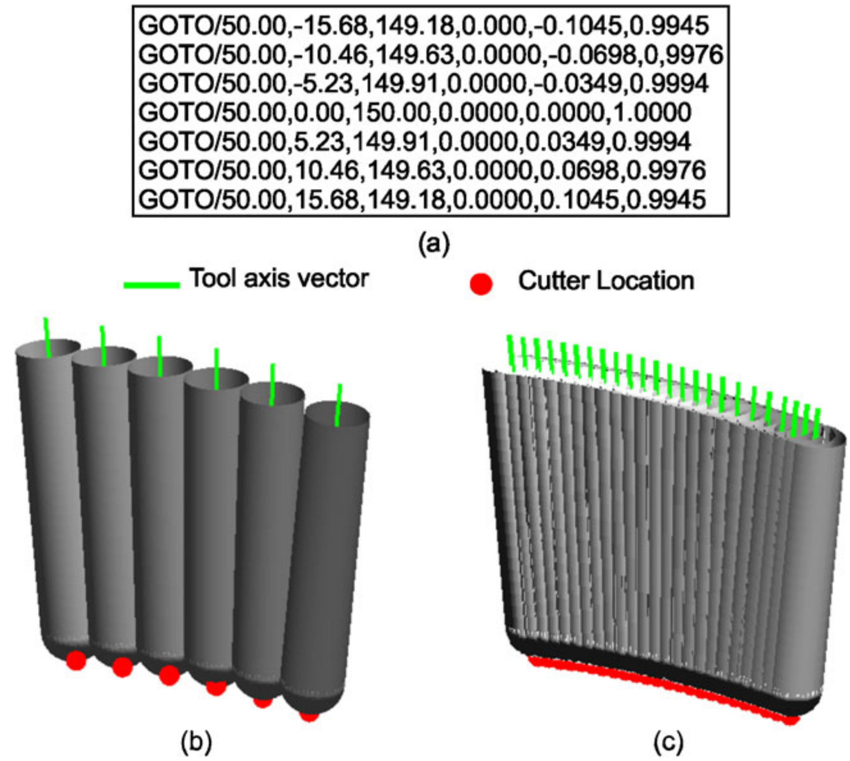


Fig. 5 Spherical interpolation, **a** example CL data, **b** original tool movement, **c** interpolated tool movement

Modification in C axis angles has to be made as well. In Eq. (13), C axis angle is determined by *atan2* function which is a four quadrant function and gives the rotation angles in the range of [−180°, 180°]. Especially around kinematic singularities, C axis of the machine may make quick turns [8] which may damage the workpiece or may even result in collision. Kinematic singularities occur if primary rotary axis angle is equal or close to zero (θ_y=0) where tool axis is parallel to the secondary rotary axis of the machine (C axis). At this instant, the rotation angle for the secondary rotary axis is undefined according to [8].

Quick turns along the toolpath may also be observed, if the actual operating range of the machine (e.g. [0°, 360°]) allows more or less than the calculation domain and calculated axis positions exceed these limits an alternative solution is generated. If this is the case, rotary axis makes a half or full rotation in the opposite direction to reach the commanded position. For instance, assume that the current C axis angle is 180° and the following C axis angle on the machine tool is 185°; however, calculated C axis angle will be −175° due to calculation domain of *atan2* function.

In the proposed approach, shown in Fig. 7, previous C axis angle $C_{i\text{-}1}$ is compared with the current C axis angle C_i *calculated* for each CL point in the toolpath if

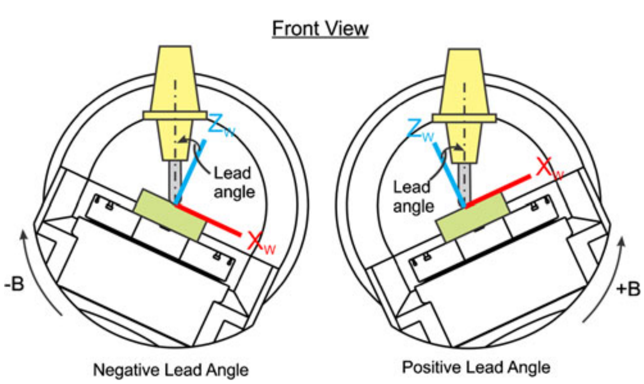


Fig. 6 Relationship between lead angle and B axis angle

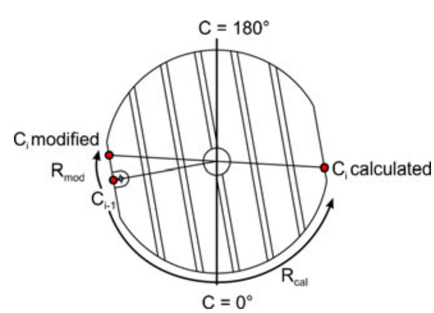


Fig. 7 C axis angle modification method

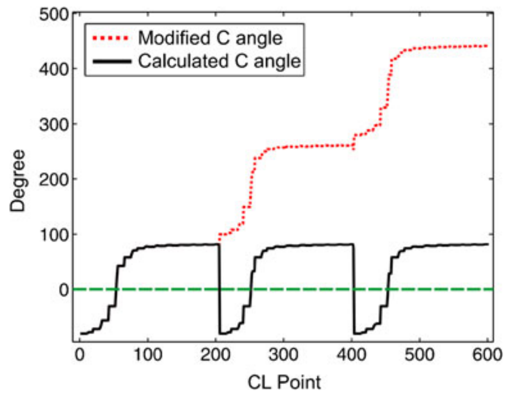


Fig. 8 C axis angle modification example

the absolute difference between two consecutive rotations is greater than 90°; calculated rotation angle is increased or decreased by ±90° until the modified rotation angle and the previous rotation angle lie in the appropriate range.

An example of the proposed approach is shown in Fig. 8. From Fig. 8, it is obviously seen that with the modification approach, quick turns along the toolpath are avoided and a continuous rotation around the rotary axis is obtained. Since continuous rotation in the toolpath along the same feed direction is satisfied, possible feed marks due to feed direction change are also eliminated with this approach.

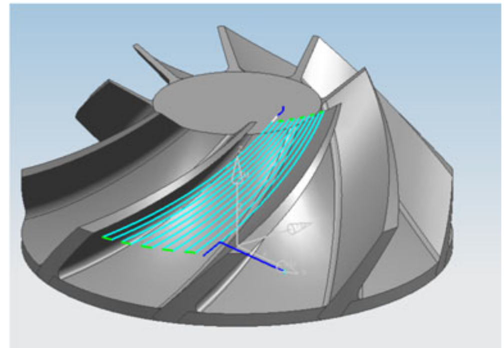
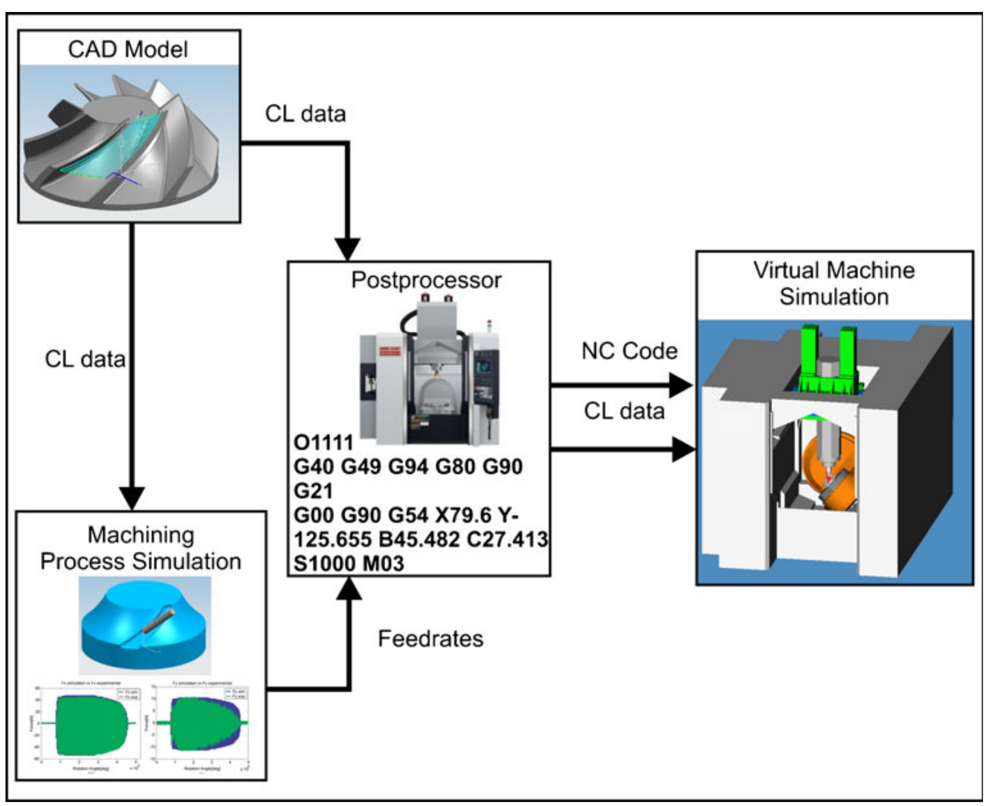


Fig. 10 Impeller roughing toolpath

4 Virtual simulation module

A virtual machine simulation module is developed for error proofing of the postprocessed NC code with the used CL file. In conventional approach, postprocessing and virtual machine simulation is performed separately with commercially available software such as Vericut. The main aim in this software is verification of the machining operation using the stock workpiece, NC code, and the machine kinematical model considering errors such as collisions, gouges etc. Therefore, this approach does not provide a proper way of judging the postprocessor performance directly since they focus on the toolpath. The approach proposed here performs the validation of the generated NC code focusing on the

Fig. 9 Virtual simulation module flow chart



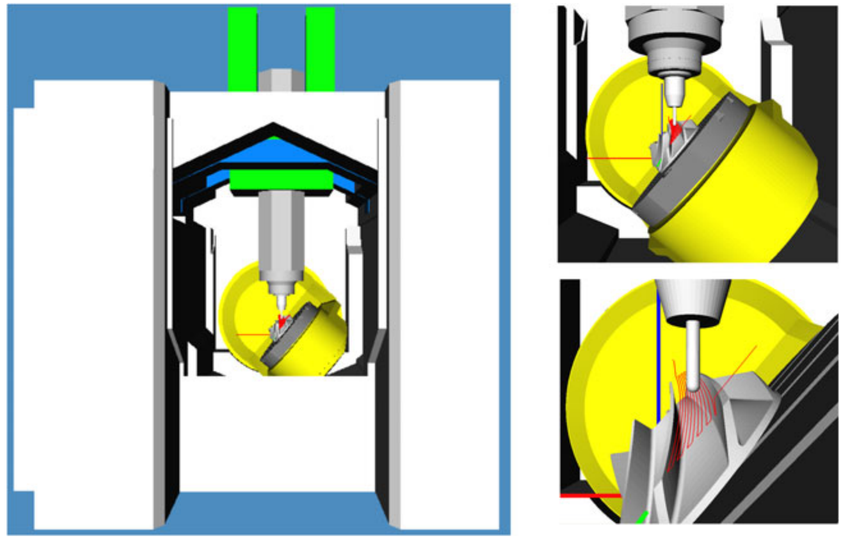


Fig. 11 Virtual machine simulation results for BC type table-tilting five-axis machine tool

machine tool kinematics. For instance, if the tool is colliding with the stock workpiece in the machining simulation, one has to check the position of the stock workpiece, generated toolpath in the CAM software, and the output NC code. However, if the main focus is only postprocessing, one has to verify that the tool tip position in the machine simulation is coincident with the CL file coordinates in the simulation.

Flow chart for virtual machine simulation module is shown in Fig. 9. The approach for virtual simulation can be summarized as follows. First, toolpath is generated using a CAM package, then the output CL data is used for physical simulation of machining process in machining process simulation module where cutting forces are simulated and a feedrate scheduling strategy is applied [10, 11]. Corresponding feedrates for each CL point is obtained. Finally, NC code is generated by using this feedrate array and the CL data.

In virtual machine simulation module, machine parts are assembled using STL (stereolithography) files for each axis of the machine. All simulation related parameters such as tool offset, workpiece zero, machine home

Fig. 12 Sample output of the NC code

```
GOTO/82.8605,-91.6376,58.1050,0.4249700,-0.4402500,0.7909400
PAINT/COLOR,211
RAPID
GOTO/72.6613,-81.0715,39.1226
PAINT/COLOR,6
FEDRAT/MMPM,250.0000
GOTO/71.8955,-79.9926,37.7908
GOTO/71.2488,-78.4982,36.8568
GOTO/70.7844,-76.7346,36.4118
GOTO/70.5479,-74.8744,36.4995
GOTO/70.5622,-73.0998,37.1113
PAINT/COLOR,31
GOTO/70.7372,-71.0301,38.1730,0.4385124,-0.4403281,0.7834654
GOTO/70.8988,-68.9653,39.2465,0.4522313,-0.4398985,0.7758712
GOTO/71.0466,-66.9053,40.3312,0.4661103,-0.4389414,0.7681612
GOTO/71.1800,-64.8503,41.4270,0.4801320,-0.4374367,0.7603436
GOTO/71.2982,-62.8003,42.5338,0.4942763,-0.4353644,0.7524286
GOTO/71.4005,-60.7553,43.6514,0.5085206,-0.4327041,0.7444287
```

Rapid motion

Engage motion

Cut motion

CL Data from the CAM software

```
G00 G90 G53 X62.11 Y-4.026 B37.727 C-46.012 S600 M03
G43.4 Z121.514 H5
Z97.514
G01 X61.89 Y-3.828 Z95.66 F250.
X61.256 Y-3.255 Z93.989
X60.269 Y-2.365 Z92.663
X59.027 Y-1.243 Z91.813
X57.651 Y0.0 Z91.522
X54.816 Z92.203 B38.421 C-45.118
X51.98 Z92.851 B39.116 C-44.208
X49.143 Z93.464 B39.811 C-43.281
X46.307 Z94.042 B40.506 C-42.336
X43.474 Z94.586 B41.199 C-41.374
X40.645 Z95.093 B41.89 C-40.395
```

NX6 Post Builder Generated

```
G00 G90 G54 X62.111 Y-4.027 B37.727 C-46.012 S600 M03
G43 Z121.514 H5
X62.110 Z97.514
G01 X61.891 Y-3.828 Z95.660 F500
X61.256 Y-3.256 Z93.988 F500
X60.270 Y-2.365 Z92.663 F500
X59.028 Y-1.243 Z91.813 F500
X57.652 Y-0.000 Z91.521 F500
X54.816 Z92.203 B38.421 C-45.118 F321
X51.980 Z92.851 B39.116 C-44.208 F321
X49.143 Z93.464 B39.811 C-43.281 F321
X46.307 Z94.042 B40.506 C-42.336 F321
X43.474 Z94.586 B41.199 C-41.374 F321
X40.645 Z95.093 B41.890 C-40.395 F321
```

Engage feedrate

Cut feedrate

Proposed method

coordinate are set. NC code is used for the motion simulation of the machine axes which are postprocessed relative to workpiece zero. CL data is used for the illustration of the toolpath in the simulation. Validity of the NC code is proved by comparing CL data (toolpath in simulation) and tooltip position of the machine. If tooltip position and the corresponding CL point match, then NC code is verified.

5 Implementation and verification

The virtual machine simulation module described in Section 4 is implemented in Visual Studio C++ using the Coin3D graphics development API. Implementation of the proposed approach and virtual machine simulation module is verified on an impeller roughing toolpath which is shown in Fig. 10. Toolpath is generated in Siemens NX6 CAM software.

For postprocessing, programmed zero is located at the bottom center of the workpiece and part is directly mounted on the C axis table, therefore workpiece offset distance is set to $W = 0i + 0j + 80k$ mm. Selected tool and the holder have a tool offset length of 150 mm. The distance from the machine home position to the workpiece zero is 365$\boldsymbol{i}$−255$\boldsymbol{j}$−640$\boldsymbol{k}$ mm. Virtual machine simulation module user interface and close-ups for 300th CL points are shown in Fig. 11. In Fig. 12, sample output of the NC code and the code postprocessed with a commercial postprocessor generated by NX6 Post Builder is shown.

From Figs. 11 and 12, it can be seen that programmed five-axis motion in the NC code matches perfectly with the CL data positions; therefore, it may be concluded that the validity of the presented postprocessor and virtual machine simulation module is proved.

A second verification test is performed on an AC-type table-tilting five-axis machine tool in order to demonstrate that presented approach is generic. A multi blade milling toolpath for impeller geometry is generated using NX 7.5 CAM software. Generated toolpath consists of 13,108 CL points. Details of the toolpath are illustrated in Fig. 13.

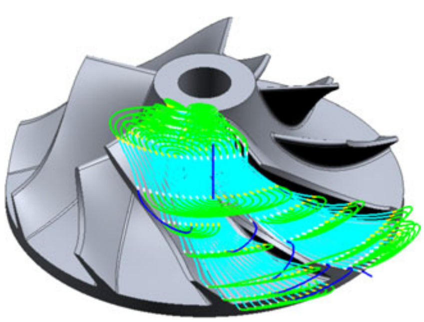


Fig. 13 Multi blade milling toolpath

In the verification test, a machining scenario is built where the bottom of the blank workpiece is located at the machine rotary zero position. Work offset (programmed zero) is located at the rotary zero of the machine, therefore workpiece offset distance is set to $W = 0i + 0j + 0k$ mm. Selected tool is a 12-mm diameter ball-end mill and the tool offset length is 150 mm from the tooltip of the cutter to the spindle. The distance from the machine home position to the workpiece zero is −315$\boldsymbol{i}$+0$\boldsymbol{j}$−605$\boldsymbol{k}$ mm. Screenshots of virtual machine simulation module and close-ups are shown in Fig. 14 while the simulation is running. The results of the second simulation demonstrate that generated NC data matches with the CL data (toolpath) for machine tool with AC type table-tilting machine configuration as well.

6 Conclusion

In this paper, a postprocessor for table-tilting/rotating type five-axis machine tool based on generalized kinematics and a virtual machine simulation module are presented. The approach developed based on this model is modular. Therefore, machine tools with different kinematic configuration can be incorporated into the model very easily. The approach has the ability to generate the NC code automatically for a given CL file with its verification.

The presented method can be used in industry, for postprocessor development and it can be integrated into CAD/CAM programs. Moreover, it may provide a safe and cost-effective solution for the multi-axis virtual machine simulation.

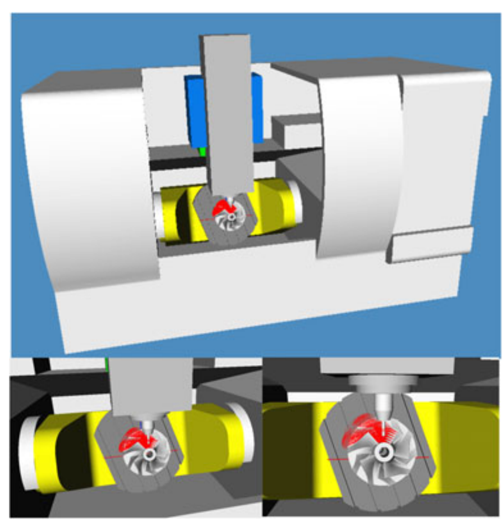


Fig. 14 Virtual machine simulation results for AC type table-tilting five-axis machine tool

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